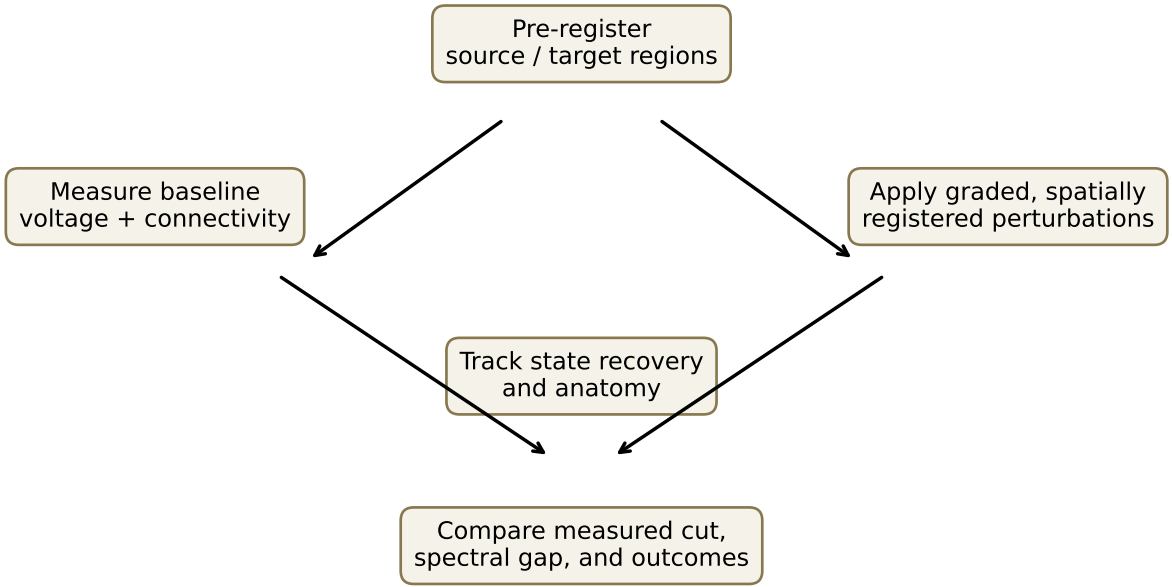Prospective biological validation plan (not yet performed)